

Worked Example: Kota Padang

Step-by-step calculation of climate impact, cost, fossil energy and land take per tonne of MSW, with uncertainty and sensitivity analysis — MSW pathway model v3.1, baseline year 2025

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Summary for Kota Padang (central values unless stated)

- 2025 tonnage 656.0 t/day; composition sampled in 2023 and used as a 2025 proxy; bulk moisture 0.52; LHV 7.19 MJ/kg; methane potential 45.8 kg CH₄/t (MCF = 1); fossil CO₂ if burned 434 kg/t.
- GHG per tonne: open dump 990, sanitary landfill 561, WtE 121, RDF 350, AD 371, PHB 547, RDF + AD 164 kg CO₂e/t.
- Net cost per tonne: landfill 18.4, WtE 57.9, RDF 30.2, AD 30.6, PHB 38.1, RDF + AD 32.9 USD/t.
- At a carbon value of 50 USD/t the lowest social cost is S5 RDF + AD; RDF + AD overtakes the landfill at 37 USD/t CO₂e.
- Monte Carlo (4,000 draws, carbon value 0–100 USD/t): RDF + AD is best in 49% of draws and the landfill in 46%: a statistical tie.
- Sensitivity: as-received moisture and landfill-gas collection efficiency dominate the comparison between landfill and RDF + AD (Sobol total indices 0.33 and 0.21).

How this document was produced. *docs/padang_calc.py* writes out every equation of the model again with named intermediate quantities and stops if any result differs from the model output (GHG, cost, CED and land take of all options agree to 10⁻⁶). *docs/padang_uncertainty.py* runs the Monte Carlo with the same random numbers as the 21-location analysis, so the probabilities are identical to Table 7 of the methodology. Symbols follow the methodology (Section 4.1).

1 Inputs from the RIPS

Source: RIPS extract (author); RIPS Kota Padang (komposisi domestik/non-domestik terpisah). Composition sampled in 2023 (wet mass, domestic and non-domestic reported separately). Glass is not reported for Padang: the cell is blank and treated as not reported, not as zero. Data-quality flags: C4 glass not reported (blank, not zero); C0 composition is the RIPS sampling-year distribution used as a 2025 proxy, not a 2025 measurement

RIPS category	Domestic %	Non-domestic %	Model fraction
sisa_makanan	43.95	51.56	food
daun_kering	15.68	25.78	garden
kertas_kardus	8.85	6.34	paper
kayu	0.42	0.26	wood
plastik	19.84	9.61	plastic
logam	1.30	1.36	metal
kain	1.80	0.93	textile
karet_kulit	0.27	0.26	rubber
sterofoam	0.22	0.54	plastic
b3	0.18	0.33	other
elektronik	1.09	0.17	other
lain_lain	6.41	2.85	other
kaca	blank	blank	glass
Sum	100.01	99.99	

Table 1: RIPS composition of Kota Padang and its mapping to the ten model fractions.

Tonnage to 2025. The tonnage comes from the RIPS table of the stated year, so it is projected once with the Padang growth rate:

$$Q_{2025} = Q_t(1 + g)^{2025 - t} = 647.57 \times (1 + 0.0065)^2 = 656.02 \text{ t/day} \quad (1)$$

Domestic 620.73 → 628.83 t/day and non-domestic 26.84 → 27.19 t/day (same factor 1.013042). The tonnage does not enter the GHG per tonne; it enters plant cost (economies of scale) and the scale gates.

2 Composition per functional unit

Each stream is normalised to 1 (domestic sum 100.01%, factor 0.99990; non-domestic 99.99%, factor 1.00010) and the two are weighted by their 2025 masses (95.86% / 4.14%):

$$S_j = \frac{M_d S_j^{(d)} + M_n S_j^{(n)}}{M_d + M_n} \Rightarrow \sum_j S_j = 1 \text{ t per t MSW} \quad (2)$$

Fraction	Domestic (mapped %)	Non-domestic (mapped %)	s domestic	s non-domestic	s used (t/t)
food	43.95	51.56	0.4395	0.5157	0.4426
garden	15.68	25.78	0.1568	0.2578	0.1610
paper	8.85	6.34	0.0885	0.0634	0.0875
wood	0.42	0.26	0.0042	0.0026	0.0041
textile	1.80	0.93	0.0180	0.0093	0.0176
rubber	0.27	0.26	0.0027	0.0026	0.0027
plastic	20.06	10.15	0.2006	0.1015	0.1965
metal	1.30	1.36	0.0130	0.0136	0.0130
glass	0.00	0.00	0.0000	0.0000	0.0000
other	7.68	3.35	0.0768	0.0335	0.0750

Table 2: Composition per tonne of MSW. Rules L and W are not triggered for Padang (organics reported separately; dry leaves present).

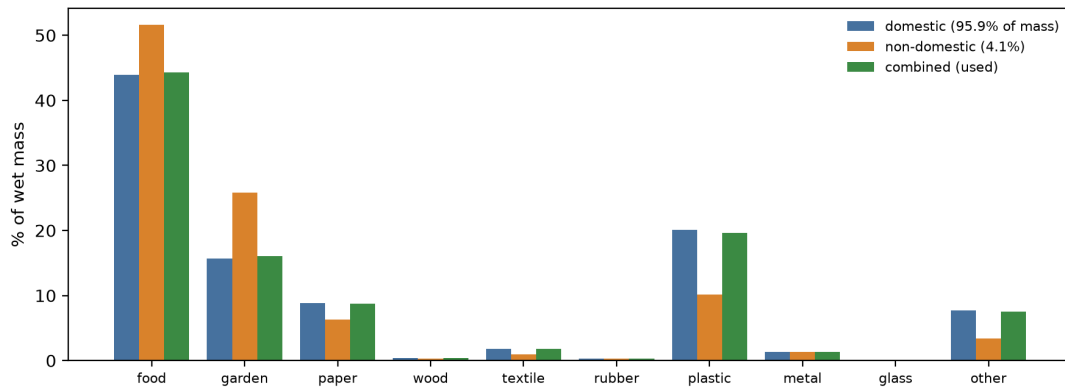


Figure 1: Domestic, non-domestic and combined composition of Kota Padang.

3 Characterisation: dry matter, methane potential, fossil carbon, heating value

Every property multiplies the same dry mass $s_j(1 - w_j)$ (moisture as received, main case). Parameters: Table 2 of the methodology; $\text{DOC} \times \text{DOCf}$ multiplier $k_D = 1.00$, methane fraction $F = 0.50$, heating-value factor $k_h = 0.90$ (legacy fractions; plastic uses its own value), rubber $\text{DOCf} = 0$ (gap scenario).

$$\text{CH}_4 = 10^3 \sum_j S_j (1 - w_j) \text{DOC}_j \text{DOCf}_j k_D F \frac{16}{12}; \quad E_{\text{fos}} = 10^3 \sum_j S_j (1 - w_j) C_j \phi_j \frac{44}{12}; \quad H = \sum_j S_j (1 - w_j) h_j - \lambda \sum_j S_j w_j \quad (3)$$

Fraction	s	w	dry s(1-w)	DOC	DOCf	C	ϕ	h MJ/kg	CH4 kg/t	CO2 fossil kg/t	LHV MJ/kg
food	0.4426	0.75	0.1107	0.38	0.70	0.38	0.00	14.85	19.62	0.00	0.832
garden	0.1610	0.55	0.0724	0.49	0.70	0.49	0.00	14.40	16.56	0.00	0.827
paper	0.0875	0.35	0.0568	0.44	0.50	0.46	0.01	14.40	8.34	0.96	0.744
wood	0.0041	0.25	0.0031	0.50	0.10	0.50	0.00	16.20	0.10	0.00	0.048
textile	0.0176	0.35	0.0115	0.30	0.50	0.50	0.20	16.65	1.15	4.20	0.176
rubber	0.0027	0.15	0.0023	0.47	0.00	0.67	0.20	20.70	0.00	1.13	0.046
plastic	0.1965	0.22	0.1532	0.00	0.00	0.75	1.00	30.50	0.00	421.44	4.569
metal	0.0130	0.05	0.0124	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.002
glass	0.0000	0.02	0.0000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.000
other	0.0750	0.25	0.0562	0.00	0.00	0.03	1.00	0.00	0.00	6.19	-0.046
Total	1.0000	M = 0.521	0.4787						45.77	433.91	7.194

Table 3: Characterisation per tonne of MSW. The LHV column is each fraction's contribution $s(1-w)h - \lambda sw$ (water makes it negative for fractions without heating value).

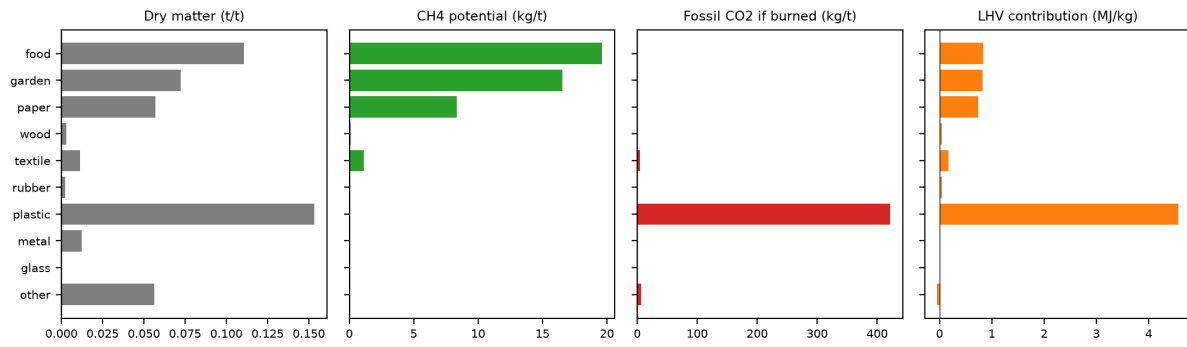


Figure 2: Contribution of each fraction to dry matter, methane potential, fossil CO₂ and heating value.

Food is 44% of the wet mass but, being wet ($w = 0.75$), only 0.111 t/t of dry matter; garden waste (16% of the mass, $w = 0.55$, DOC 0.49) contributes almost as much methane (16.6 vs 19.6 kg/t). Plastic supplies 97% of the fossil CO₂ and 4.57 of the 7.19 MJ/kg. The dry-matter HHV implied is 19.0 MJ/kg, inside the 15.7–19.2 MJ/kg measured for Indonesian MSW (Prabowo et al. 2019).

4 Baselines: open dump and sanitary landfill

Open dump (MCF = 0.8, no gas collection, no oxidation): $CH_4 = 45.77 \times 0.8 = 36.62$ kg/t; $G_{OD} = 36.62 \times 27 + 1$ (diesel) = **989.7 kg CO₂e/t**; cost 4.0 USD/t.

Sanitary landfill (MCF = 1, lifetime gas collection $\eta = 0.50$, flare, cover oxidation 0.10): CH_4 generated 45.77 kg/t, captured and flared 22.89, emitted $(45.77 - 22.89) \times (1 - 0.10) = 20.60$ kg/t; $G_{SL} = 20.60 \times 27 + 5 =$ **561.1 kg CO₂e/t**. Cost: $c_{SL} (Q/500)^{-e} = 20 \times (656.0/500)^{-0.3} =$ **18.44 USD/t**.

The landfill already avoids 429 kg CO₂e/t relative to the open dump, for 14.4 USD/t more.

5 The five recovery options, step by step

Common values: grid Sumatera EF = 0.94 kg CO₂/kWh; nearest kiln Semen Padang (Indarung) at 16.8 km by road; $CRF = r(1+r)^n / ((1+r)^n - 1) = 0.1175$ ($r = 0.10$, $n = 20$); CAPEX per tonne = $K_{ref} (q/0.85/q_{ref})^b CRF / (365q)$ with $b = 0.70$.

5.1 S1 Waste-to-energy

Electricity $E = H \times 1000 / 3.6 \times \eta_w = 7.194 \times 277.8 \times 0.18 = 359.7$ kWh/t.

GHG component	kg CO ₂ e/t	Cost component	USD/t
fossil	433.91	capex	49.10
N ₂ O	13.65	om	27.00
auxiliary	10.00	revenue	-25.18
electricity credit	-338.11	ash landfill	6.99
ash landfill	1.25		
Total G	120.70	Total C	57.91

Table 4: WtE. Fossil CO₂ from Table 3; N₂O 0.05 kg/t $\times$ 273; ash 0.25 t/t to landfill (inert); electricity credited at the grid factor and sold at 0.07 USD/kWh (market case). Gate G1 (LHV ≥ 7 MJ/kg): passed; PSEL tariff ($\geq 1,000$ t/day): not eligible.

5.2 S2 RDF to cement kiln

Fraction	s (t/t)	τ (x k)	to RDF r (t/t)	rejected (t/t)
food	0.4426	0.10	0.0443	0.3984
garden	0.1610	0.40	0.0644	0.0966
paper	0.0875	0.70	0.0612	0.0262
wood	0.0041	0.60	0.0025	0.0017
textile	0.0176	0.70	0.0123	0.0053
rubber	0.0027	0.60	0.0016	0.0011
plastic	0.1965	0.80	0.1572	0.0393
metal	0.0130	0.00	0.0000	0.0130
glass	0.0000	0.00	0.0000	0.0000
other	0.0750	0.10	0.0075	0.0675

Table 5: Sorting of each fraction into RDF (legacy transfer coefficients).

RDF line: mass in 0.3510 t/t with 0.1317 t water and 0.2193 t dry; dried to $\omega = 0.20$: $m_{\text{out}} = \min(0.3510, 0.2193/(1-0.20)) = 0.2741$ t/t. Energy in dried RDF $E = \Sigma r(1-w)h - \lambda(m_{\text{out}} - \text{dry}) = 4.952$ GJ/t; dryer heat (3.2 GJ/t water $\times$ 0.0768 t) leaves $E_{\text{net}} = 4.706$ GJ/t and $m_{\text{del}} = 0.2605$ t RDF/t MSW with NCV = 18.06 MJ/kg (gate G4 $\geq$ 12.56: passed). Coal displaced: $\psi E_{\text{net}} = 0.90 \times 4.706$ GJ at 96.1 kg CO₂/GJ. The rejects (0.6490 t/t, mostly wet food) are landfilled and generate 30.49 kg CH₄/t.

GHG component	kg CO2e/t	Cost component	USD/t
fossil	342.06	capex	11.69
electricity	37.60	om	12.00
diesel	3.00	transport	0.44
transport	0.44	revenue	-7.53
coal credit	-407.04	rejects landfill	13.62
rejects landfill	373.64		
Total G	349.70	Total C	30.22

Table 6: RDF. The methane from rejected wet organics is the largest positive term.

5.3 S3 Anaerobic digestion of food waste

Food captured $a = \kappa s_{\text{food}} = 0.70 \times 0.4426 = 0.3098$ t/t. Methane $V = a \times 1000 \times (1-w_{\text{food}}) \times VS/TS \times y_{\text{CH}_4} = 0.3098 \times 1000 \times 0.25 \times 0.87 \times 0.36 = 24.26$ Nm³/t; fugitive 0.05; electricity exported 66.2 kWh/t (η_{CHP} 0.36, own use 0.20). The rest of the waste (0.6902 t) and the digestate (0.0620 t) go to the landfill (32.04 kg CH₄/t generated). Front-end separation of food from mixed waste is charged as $\pi = 0.40$ of the RDF line.

GHG component	kg CO2e/t	Cost component	USD/t
fugitive	23.45	capex	6.32
diesel	0.93	om	4.34
electricity credit	-62.22	revenue	-4.63
residue landfill	393.02	residue landfill	15.10
front end sorting	16.24	front end sorting	9.48
Total G	371.42	Total C	30.60

Table 7: AD of food waste separated from mixed waste.

5.4 S4 PHB from landfill gas

Captured methane 22.89 kg/t $\div R = 2.3$ t CH₄/t PHB gives 9.95 kg PHB/t, i.e. 2,383 t PHB/yr at 656 t/day (gate G6 $\geq$ 500 t/yr: passed). Production cost $c_{500} (P/500)^{-e} = 5.97$ USD/kg against a price of 4.0 USD/kg.

GHG component	kg CO2e/t	Cost component	USD/t
landfill CH4	556.15	landfill	18.44
diesel	5.00	PHB production	59.44
PHB chemicals	4.98	PHB revenue	-39.80
PP credit	-18.91		
Total G	547.22	Total C	38.07

Table 8: PHB. Against the landfill it is built on, the only climate gain is the displaced polypropylene.

5.5 S5 Integrated RDF + AD

The whole tonne passes the RDF line, which also separates the food: 0.3098 t to the digester (as in S3), the remaining 0.6902 t sorted as in S2 (RDF 0.2548 t/t, NCV 18.31 MJ/kg); only the rejects (0.3702 t/t) and digestate go to the landfill (18.12 kg CH₄/t).

GHG component	kg CO2e/t	Cost component	USD/t
AD fugitive	23.45	AD capex	6.32
AD diesel	0.93	AD om	4.34
AD electricity credit	-62.22	AD revenue	-4.63
RDF fossil	342.06	RDF capex	11.69
RDF electricity	37.60	RDF om	12.00
RDF diesel	3.00	RDF transport	0.43
RDF transport	0.43	RDF revenue	-7.46
RDF coal credit	-403.39	rejects landfill	10.25
rejects landfill	222.35		
Total G	164.20	Total C	32.93

Table 9: RDF + AD. No separate front-end step is charged because the RDF line separates the food.

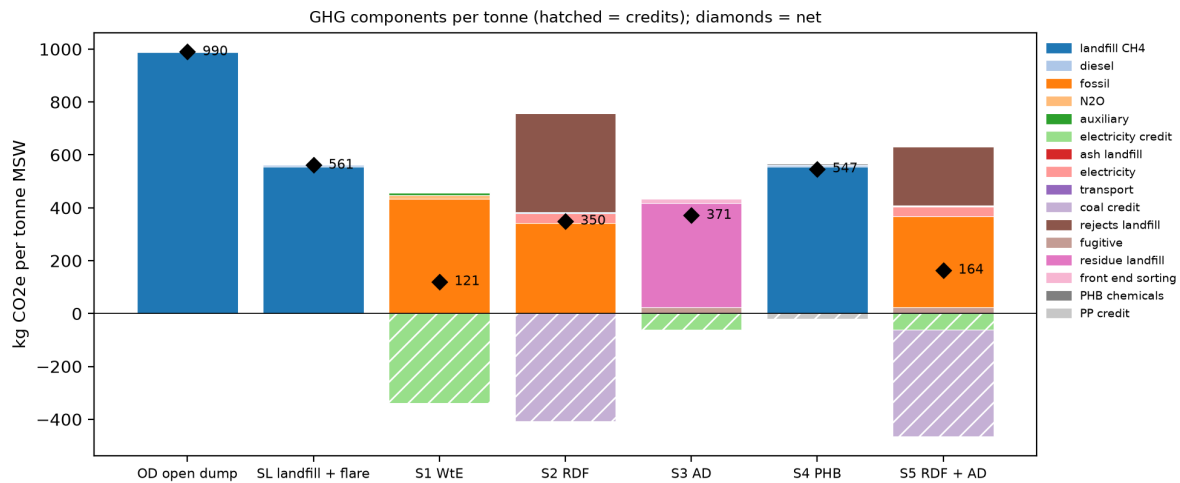


Figure 3: GHG components of each option for Kota Padang.

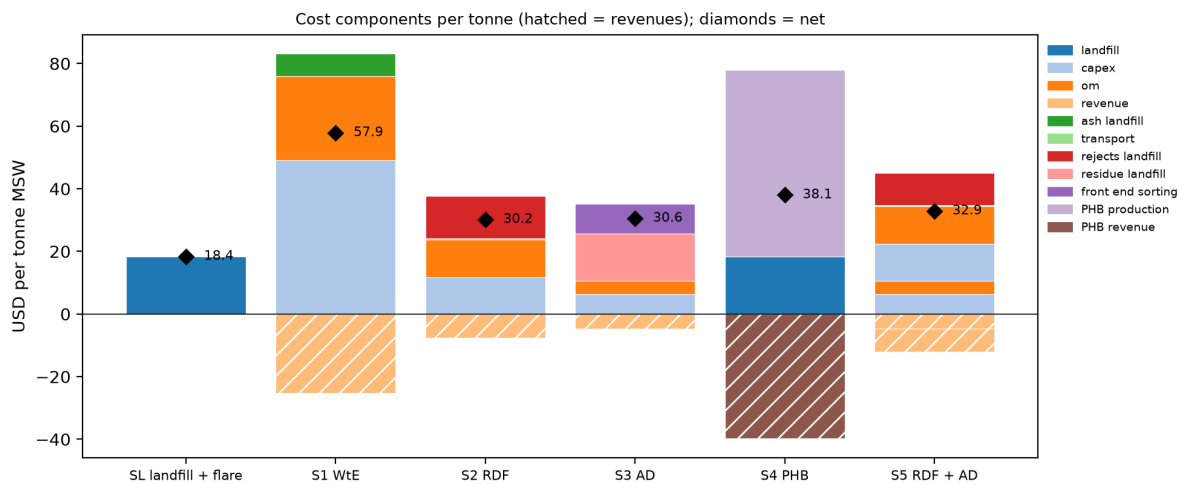


Figure 4: Cost components of each option for Kota Padang.

6 Summary of central results and decision

Option	Gates	G kg CO2e/t	ΔG vs OD	ΔG vs SL	C USD/t	MAC vs SL USD/t	CED MJ/t	Land m2/t
Open dump (baseline)	-	989.7	-	-	4.0	-	-	-
SL landfill + flare	pass	561.1	428.6	-	18.44	-	78	0.0812
S1 WtE	pass	120.7	869.0	440.4	57.91	90	-4,196	0.0189
S2 RDF	pass	349.7	640.0	211.5	30.22	56	-3,858	0.0608
S3 AD	pass	371.4	618.3	189.7	30.60	64	-518	0.0654
S4 PHB	pass	547.2	442.5	13.9	38.07	1410	-580	0.0812
S5 RDF + AD	pass	164.2	825.5	396.9	32.93	37	-4,625	0.0442

Table 10: Central results for Kota Padang (market case, GWP100).

The best option minimises the social cost $SC_k = C_k + p G_k / 1000$ among the options that pass their gates. Two options with straight lines cross at $p^* = 1000(C_b - C_a) / (G_a - G_b)$: for RDF + AD against the landfill $p^* = 1000 \times (32.93 - 18.44) / (561.1 - 164.2) = 36.5$ USD/t CO₂e.

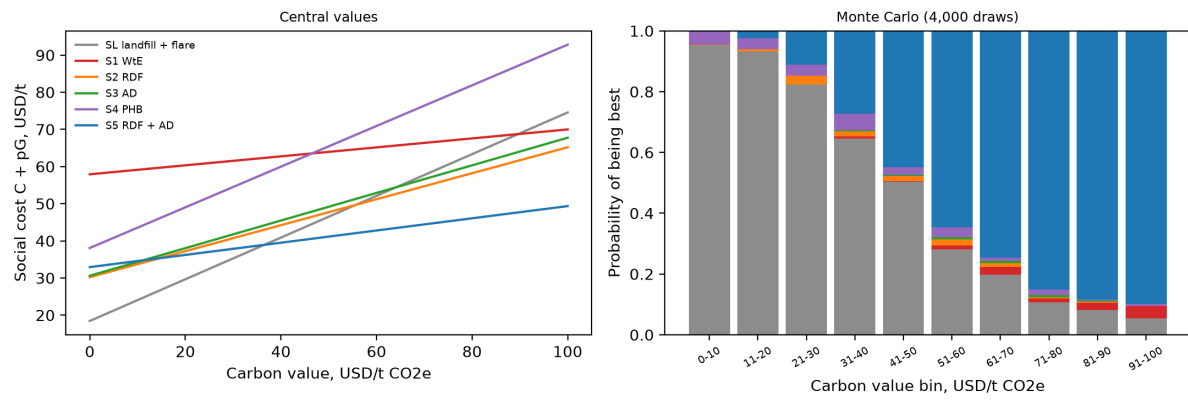


Figure 5: Left: social cost of each option against the carbon value (central values). Right: probability of being best within each carbon-value bin (Monte Carlo).

7 Uncertainty analysis

4,000 Monte Carlo draws. Composition: Dirichlet around the RIPS proxy with concentration $\alpha_0 = 80$ (heuristic; unflagged data). Parameters: triangular distributions of the register (58 inputs) and the CED/land-use factors; one common wetness draw moves all moistures; the carbon value is uniform on 0–100 USD/t.

Fraction	Central	p5	median	p95
food	0.443	0.352	0.441	0.532
garden	0.161	0.099	0.159	0.231
paper	0.087	0.042	0.085	0.145
wood	0.004	0.000	0.001	0.018
textile	0.018	0.002	0.014	0.046
rubber	0.003	0.000	0.000	0.014
plastic	0.196	0.130	0.193	0.274
metal	0.013	0.001	0.009	0.037
glass	0.000	0.000	0.000	0.000
other	0.075	0.034	0.071	0.130

Table 11: Sampled composition (share of wet mass). Bulk moisture p5–p95 0.42–0.60; LHV 5.4–9.8 MJ/kg; WtE heating-value gate passed in 60% of draws.

Option	G p5 / p50 / p95	ΔG vs OD p50 (p5–p95)	C p5 / p50 / p95	CED p50 MJ/t	P(feasible)	P(Pareto)	P(best)
SL landfill + flare	323 / 556 / 845	287 (-1-611)	13.6 / 19.0 / 24.7	87	1.00	0.95	0.461
S1 WtE	29 / 141 / 272	703 (366-1100)	44.4 / 63.4 / 85.2	-3,923	0.60	0.31	0.013
S2 RDF	183 / 344 / 528	505 (214-842)	24.2 / 31.9 / 39.8	-3,799	1.00	0.87	0.011
S3 AD	201 / 370 / 593	474 (198-774)	27.2 / 34.4 / 42.1	-444	1.00	0.44	0.003
S4 PHB	307 / 541 / 829	301 (11-634)	18.6 / 38.5 / 62.9	-487	1.00	0.28	0.027
S5 RDF + AD	53 / 163 / 288	684 (396-1021)	27.8 / 36.6 / 45.9	-4,543	1.00	1.00	0.485

Table 12: Monte Carlo results for Kota Padang (market case). $P(\text{best})$ is evaluated over the sampled carbon value.

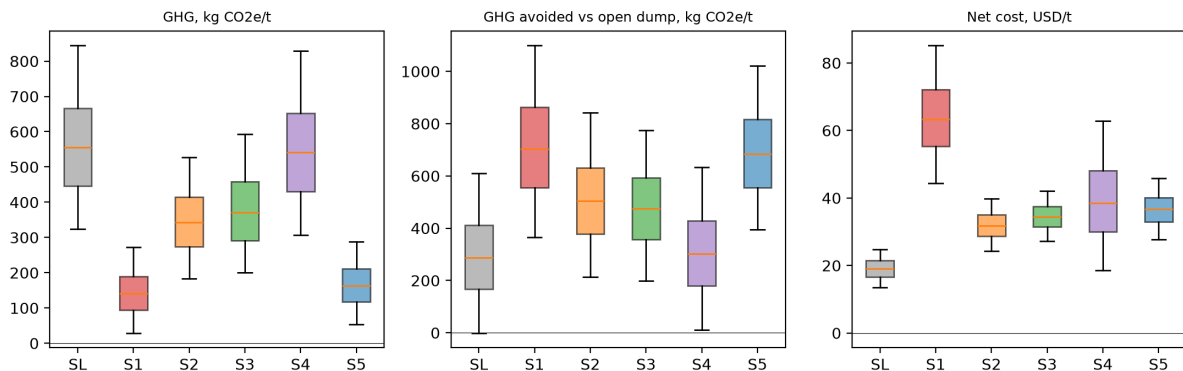


Figure 6: Distributions of GHG, GHG avoided against the open dump and net cost (boxes: interquartile range; whiskers: 5th-95th percentiles).

Indicator	Option	composition only	parameters only	both
C	S1 WtE	10.7	38.9	40.8
C	S2 RDF	5.5	14.8	15.6
C	S3 AD	0.3	14.8	14.9
C	S4 PHB	4.4	44.2	44.3
C	S5 RDF + AD	5.8	17.2	18.1
C	SL landfill + flare	0.0	11.2	11.2
G	S1 WtE	180.0	148.9	242.9
G	S2 RDF	95.9	331.8	344.6
G	S3 AD	186.1	348.7	391.6
G	S4 PHB	161.9	501.4	522.5
G	S5 RDF + AD	80.4	219.1	234.9
G	SL landfill + flare	166.1	492.5	521.6

Table 13: Width of the 90% interval ($p95 - p5$) when only one source of uncertainty is sampled. G in kg CO₂e/t, C in USD/t.

Composition uncertainty matters most for WtE (through the plastic share: 180 kg CO₂e/t of interval width) and much less for the cost of any option; parameter uncertainty dominates the landfill-based options, mainly through gas collection.

8 Sensitivity analysis

8.1 One-at-a-time (tornado). Each input of the register is set to its low and its high value with all others at their central values. The decisive output is the social-cost gap between RDF + AD and the landfill at 50 USD/t (base value -5.35 USD/t; negative = RDF + AD cheaper in social terms).

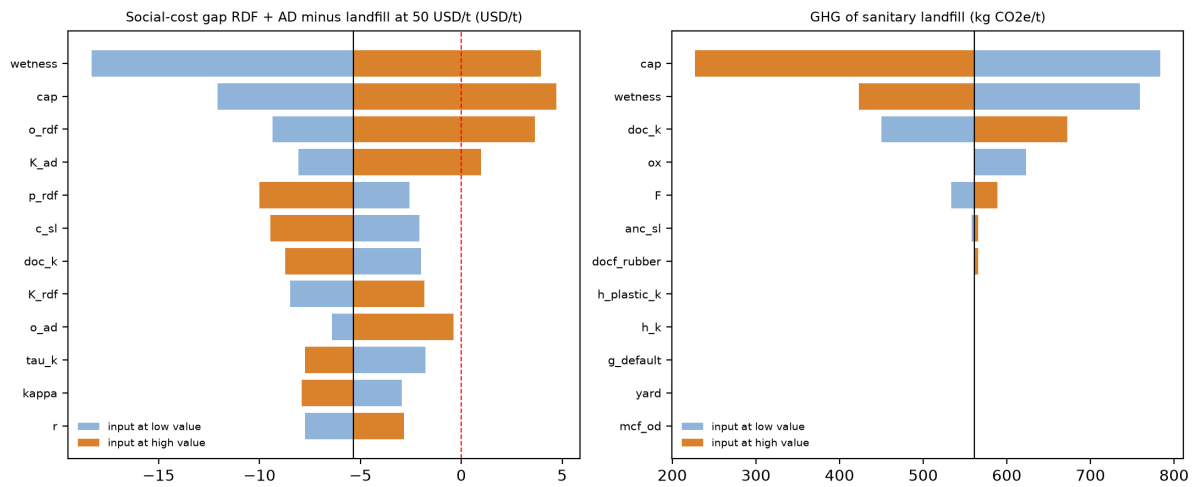


Figure 7: Tornado diagrams for Kota Padang. Left: social-cost gap S_5 minus SL at 50 USD/t (red dashed line: break-even). Right: GHG of the sanitary landfill.

Input	Meaning	Range	Gap at low	Gap at high	Crosses zero?
wetness	All fraction moistures at their low / high bound	0 to 1	-18.33	3.95	yes
cap	Lifetime landfill-gas collection efficiency, sanitary landfill	0.3 to 0.8	-12.07	4.72	yes
o_rdf	RDF plant O&M;	8 to 21	-9.35	3.65	yes
K_ad	AD plant CAPEX at reference feed capacity	4 M to 14 M	-8.06	0.97	yes
p_rdf	Price of RDF at kiln gate	1 to 2.6	-2.56	-10.02	no
c_sl	Full cost of sanitary landfill with gas collection and flare	12 to 30	-2.08	-9.45	no
doc_k	Multiplier on DOC x DOCf (default uncertainty +20%)	0.8 to 1.2	-1.99	-8.71	no
K_rdf	RDF plant CAPEX at reference capacity (Rp 200 bn)	9 M to 16 M	-8.49	-1.84	no
o_ad	AD plant O&M;	10.6 to 30	-6.41	-0.40	no
tau_k	Multiplier on RDF transfer coefficients	0.85 to 1.1	-1.78	-7.75	no

Table 14: The ten inputs with the largest swing of the $S_5 - SL$ social-cost gap. 'Crosses zero' means that the input alone, within its range, can reverse the choice between the two options.

8.2 Variance-based (Sobol). First-order (S_1) and total (ST) indices from 1,024 Saltelli base samples (all register inputs plus the wetness draw varied together; composition fixed at the Padang values).

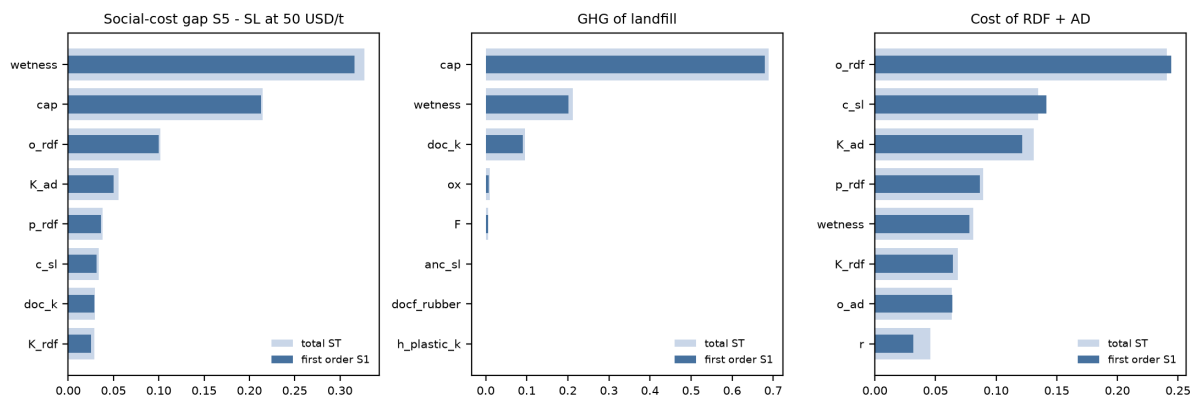


Figure 8: Sobol indices for Kota Padang.

Output	Most influential inputs: ST (S_1)
Social-cost gap $S_5 - SL$ at 50 USD/t	wetness 0.33 (0.32); cap 0.21 (0.21); o_rdf 0.10 (0.10); K_ad 0.06 (0.05); p_rdf 0.04 (0.04)
GHG landfill	cap 0.69 (0.68); wetness 0.21 (0.20); doc_k 0.10 (0.09); ox 0.01 (0.01); F 0.01 (0.01)
GHG WtE	efg_k 0.44 (0.43); eta_wte 0.43 (0.43); h_k 0.04 (0.04); wetness 0.04 (0.03); h_plastic_k 0.04 (0.04)
GHG RDF + AD	cap 0.57 (0.55); kappa 0.11 (0.10); psi 0.08 (0.08); doc_k 0.08 (0.08); h_plastic_k 0.04 (0.04)

Output	Most influential inputs: ST (S1)
Cost WtE	K_wte 0.54 (0.54); p_el 0.11 (0.10); wetness 0.09 (0.09); o_wte 0.08 (0.08); r 0.08 (0.07)
Cost RDF + AD	o_rdf 0.24 (0.24); c_sl 0.13 (0.14); K_ad 0.13 (0.12); p_rdf 0.09 (0.09); wetness 0.08 (0.08)

Table 15: Sobol indices for Kota Padang.

Option	GHG: top inputs [rho]	Cost: top inputs [rho]
SL landfill + flare	cap 0.79; moisture 0.31; doc_k 0.26; s_garden 0.20	c_sl 1.00; cap 0.05; e_sl 0.05; sub_pp 0.04
S1 WtE	s_plastic 0.74; eta_wte 0.40; efg_k 0.38; s_food 0.26	K_wte 0.69; moisture 0.32; p_el 0.31; s_plastic 0.25
S2 RDF	cap 0.80; doc_k 0.27; psi 0.17; s_food 0.17	o_rdf 0.52; c_sl 0.51; p_rdf 0.33; s_plastic 0.29
S3 AD	cap 0.72; s_garden 0.34; moisture 0.30; doc_k 0.24	c_sl 0.59; K_ad 0.40; pre 0.38; o_ad 0.27
S4 PHB	cap 0.80; moisture 0.30; doc_k 0.25; s_garden 0.19	c_phb 0.68; p_phb 0.58; c_sl 0.26; cap 0.19
S5 RDF + AD	cap 0.69; s_garden 0.29; kappa 0.28; psi 0.25	o_rdf 0.46; moisture 0.33; c_sl 0.33; K_ad 0.31

Table 16: Spearman rank correlations in the Monte Carlo sample (composition shares included as s_...).

9 Scenarios

Scenario	Q t/d	Best at 0	25	50	100	P(SL)	P(S1)	P(S2)	P(S3)	P(S4)	P(S5)
Market	656	SL	SL	S5	S5	0.46	0.01	0.01	0.00	0.03	0.49
Perpres 109	656	SL	SL	S5	S5	0.46	0.01	0.01	0.00	0.03	0.49
Food sep.	656	SL	S3	S3	S5	0.25	0.01	0.01	0.41	0.02	0.30
Residues to OD	656	SL	SL	S5	S5	0.31	0.03	0.11	0.04	0.02	0.48
Managed M1	n.a.										
Managed M2	471	SL	SL	S5	S5	0.49	0.01	0.01	0.00	0.02	0.47
GWP20	656	SL	S5	S1	S1	0.19	0.27	0.01	0.00	0.01	0.52
IPCC moisture	656	SL	S5	S5	S5	0.23	0.05	0.01	0.00	0.02	0.69
Overview projected	656	SL	SL	S5	S5	0.46	0.01	0.01	0.00	0.03	0.49
Rubber DOCf 0.5	656	SL	SL	S5	S5	0.46	0.01	0.01	0.00	0.03	0.49

Table 17: Scenarios for Kota Padang. Managed M1 is n.a. because Padang has no status-index value; M2 uses the TPA-delivery lower bound (71.8%).

Interpretation for Padang. Landfill with flare and RDF + AD are statistically tied over the carbon range; the choice turns at about 37 USD/t CO₂e at central values. A cement kiln 17 km away makes RDF cheap to deliver, and the tonnage (656 t/day) is below the Perpres 109 threshold, so the WtE tariff does not apply. Separating food at source makes AD alone the most probable option; with IPCC default moisture, RDF + AD becomes clearly preferred; with GWP20, WtE becomes the cheapest at 50–100 USD/t at central values, though RDF + AD remains most probable over the sampled range. The quantities to measure first are the as-received moisture of Padang's waste and the gas collection that a new landfill would actually achieve.

Caveats. Composition from 2023 used as a 2025 proxy; moisture as received, dry heating values and RDF transfer coefficients are assumptions; costs are class-5 estimates; the Padang managed share (93.7%) is provisional; land-use and several CED factors are assumptions.